

Maag

Modular Agentic Autonomous Greenhouse

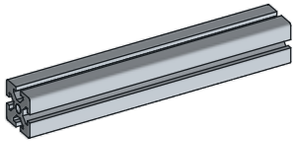
21 steps · tools · every joint

Frame and trays assembly instructions, Frame Rev A. The 4040 cabinet, 1397 × 559 × 1829, with brackets, trays and feet.

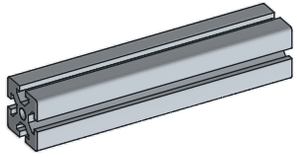
Design 0.4.0. The Maag cover render, the same on every booklet. Every drawing inside comes from cabinet.py, the Rev 4 sheet-metal outlines and the TPU foot generator.

Frame and trays · parts

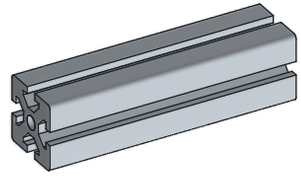
Count everything first. Quantities cover the complete cabinet.



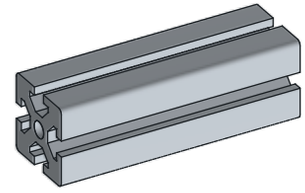
Post · 4040, 1829
4×



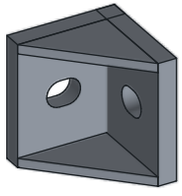
Rail · 4040, 1317
13×



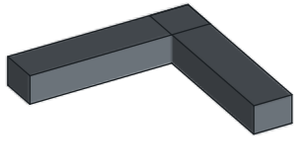
Rail · 4040, 479
11×



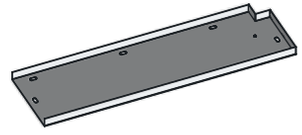
Rail · 4040, 123
3×



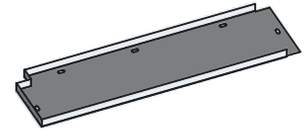
Gusset 3540, PETOX
44×



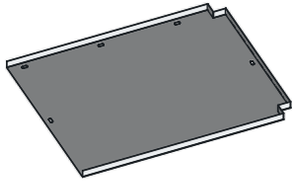
Hidden connector, TOCLUX
10×



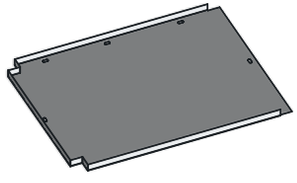
Tray, DOWN piece
3×



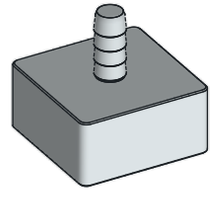
Tray, UP piece
3×



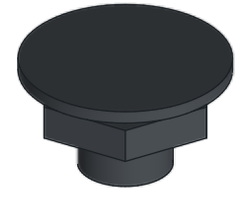
Base pan, DOWN piece
1×



Base pan, UP piece
1×



Foot · TPU 95A, printed
4×

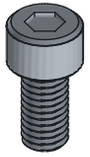


Drain bulkhead, 1/2 in
3×

4040 is EU slot 8, one slot per face. 30.075 m in all: 31 pieces from 17 sticks of 2000 (step 2). Gussets: 20 owned plus two more PETOX sets of 20, so 16 spare. Sheets: 5052-H32, 0.063 in (1.6), laser cut and bent from the Rev 4 DXFs. Feet: printed, step 4.

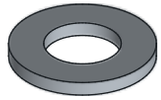
Frame and trays · fasteners

Every screw in the build. Lay them out in labelled pots first.



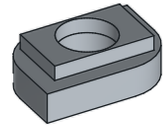
M8 × 16 cap screw · gussets

88×



M8 washer · gussets

88×



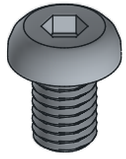
M8 drop-in T-nut · gussets

88×



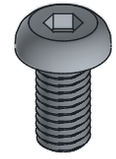
M6 grub · hidden connectors

20×



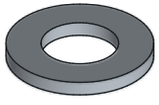
M8 × 12 button head · one sheet

28×



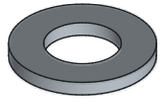
M8 × 16 button head · two layers

32×



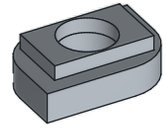
M8 washer · under sheet heads

60×



M8 washer · shims, UP pieces

28×



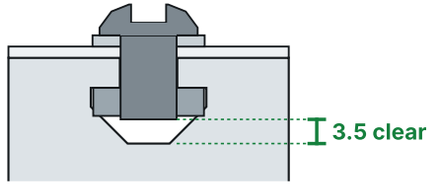
M8 drop-in T-nut · sheets

60×

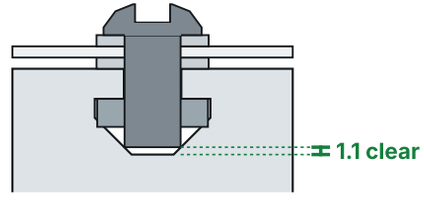
Cap screws, gusset washers and grubs come in the PETOX and TOCLUX boxes. Buy 150 drop-in T-nuts: 88 for the gussets, 60 for the sheets.

The PETOX boxes come with SLIDE-IN nuts, which only enter a slot from its end: keep them as spares. DIN 125 washers are 1.6 thick: that is the shim.

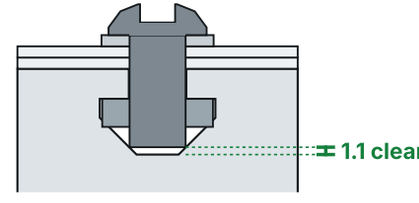
Also: neutral-cure silicone, 1 tube (steps 16 and 18); the anti-tip brackets and wall anchors, BOM G015 (step 21).



one sheet · M8 × 12



UP piece on its shim · M8 × 16

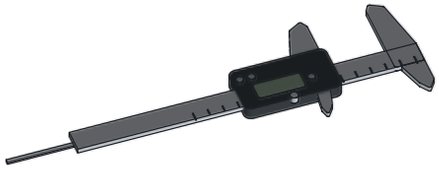


the lap, two sheets · M8 × 16

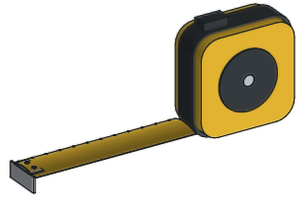
Never let a screw bottom in the slot: it feels tight and clamps nothing. On a 12.25 deep slot the M8 × 12 stops 3.5 short of the floor, the M8 × 16 through two layers 1.1 short. If step 1 measures your slot shallower, shorten both by the difference.

Tools

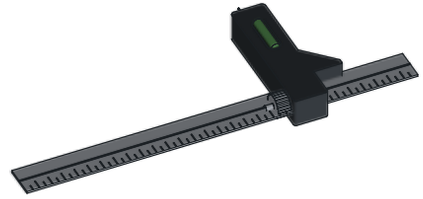
Everything the 21 steps need. Metric throughout; nothing is inch but the drain fittings.



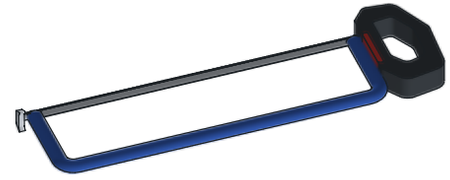
Calipers
150 mm, digital
steps: the profile, 1; the feet's bore and fit, 4



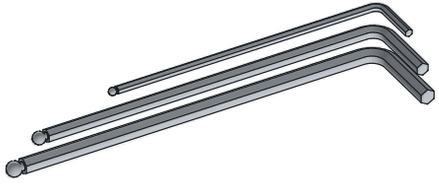
Tape measure
5 m
steps: the cut list, 2; the level marks, 3; the diagonals, 10



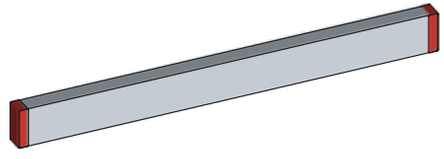
Square
300 mm combination
steps: every mark squared round the post, 3; every cut checked, 2



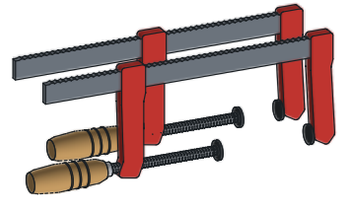
Saw
mitre saw, non-ferrous blade, or hacksaw and mitre box
steps: 31 cuts, 2. Deburr every end



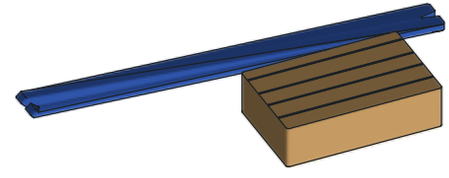
Hex keys, ball end
3 · 5 · 6 mm, long series
steps: 3: hidden connector grubs, 6 to 9
5: sheet screws, 17 to 19
6: gusset screws, 5 to 14



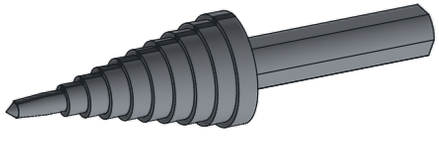
Spirit level
600 mm, with a plumb vial
steps: posts plumb, 7 and 10; the falls and the top, 21



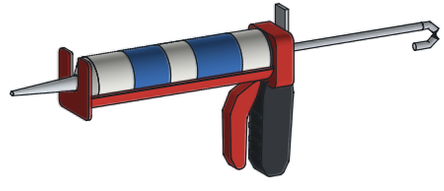
Clamps
2 F-clamps, 300
steps: a rail on its ledges while you bolt it, 9 and 12



Bar and a block
flat bar and a wood offcut
steps: lifting each corner 25 mm for its foot, 15



Step drill
to the bulkhead's hole size
steps: the Ø 6 drain pilots, 16

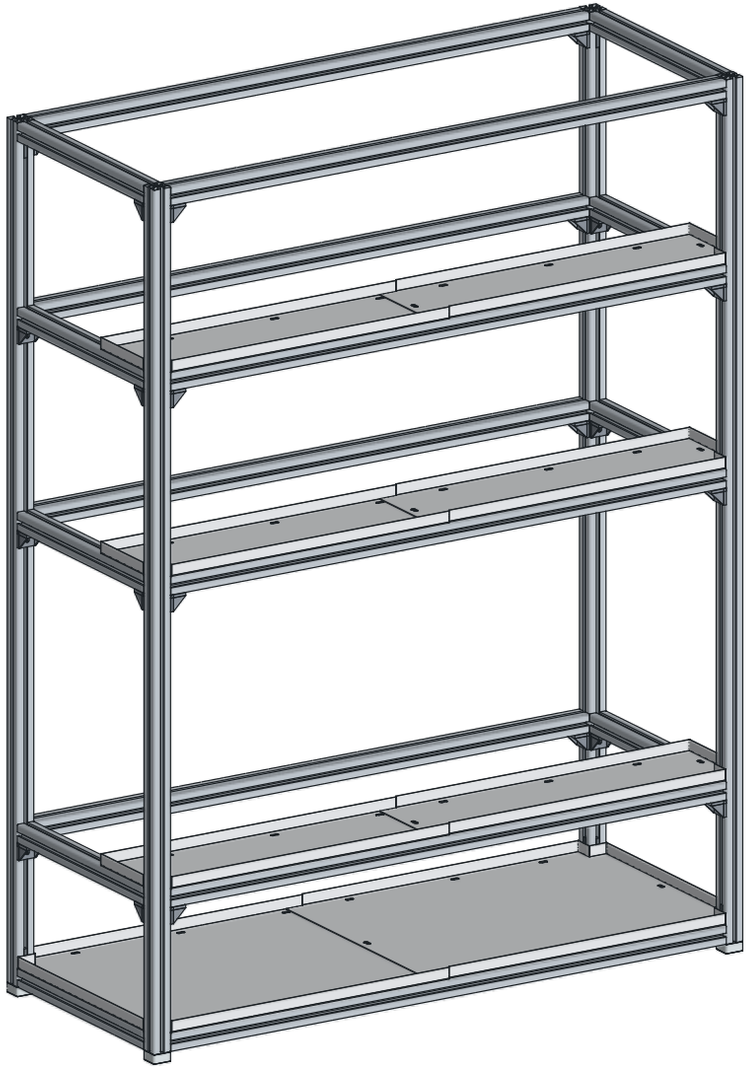


Caulking gun
and neutral-cure silicone
steps: tray corners and laps, 16 to 19

And a 3D printer for the feet (step 4), a helper for steps 7 to 9, and a few 5 mm packers for the base (step 7).

Every joint, and what holds it

54 joints, 44 gussets, 10 hidden connectors. No end taps, no drilling, any order.



Gussets

16× gusset under each front and back rail end, levels 1 to 4, leg down the post

14× gusset under each side rail end, levels 1 to 4, leg down the post

12× gusset on the front side of each shelf and tier mid rail end

2× gusset on the left side of the base mid rail, each end

Hidden connectors

8× hidden connector, every floor corner: no room below, the pan on top

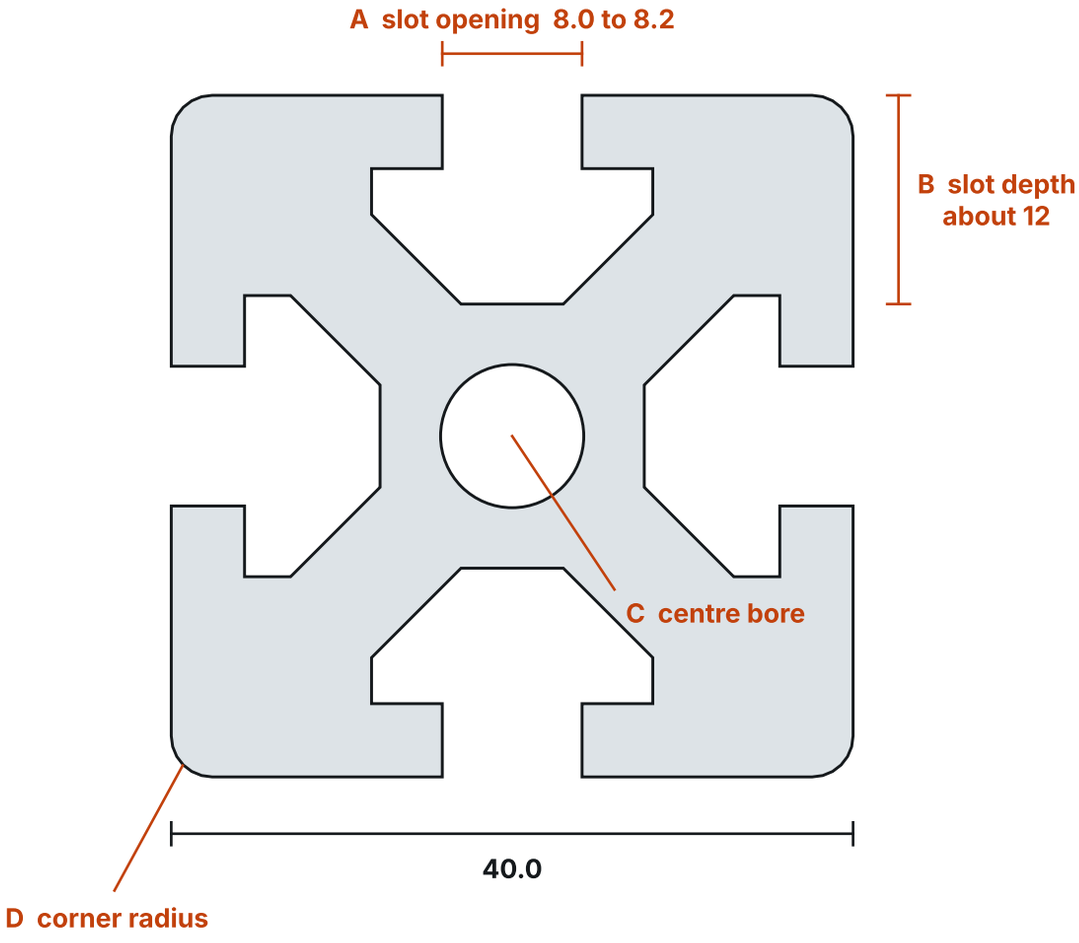
2× hidden connector, top side rails at the back: the X rail sits under them

Two rules decide every position.
No bracket on a top face: every top face is a tray seat
or the slot its nut sits in.
Nothing behind the tray line, y = 203, between the posts:
that is the gantry's corridor.

Rev 5 had an M8 end tap on every member end. Its screw head has to enter the mating slot from that slot's open end, and in this cabinet the shelf rails tie both sides together, so no order of assembly works. Brackets on drop-in nuts go on in any order, need no machining, and do not care what the profile's centre bore is.

1 Know your profile

Measure one offcut before you cut anything. Four numbers decide four things.



A Slot opening — **8.0 to 8.2**

decides whether the gussets, hidden connectors and T-nuts in this book fit. 6 or 10: stop, you have a different series

B Slot depth, face to floor — **about 12**

decides the sheet screws: shorten them by any shortfall (fasteners page)

C Centre bore — **6.8, 8.4 or 10.5**

decides which fit-test coupon and which feet to print (step 4)

D Corner radius — **about 2.4**

decides nothing structural: the feet are drawn to it. Hold a coupon against the end

`calipers`

EU slot 8, one slot per face: the brackets you own are made for it. Drawn nine times size: the generic slot-8 section, yours may have hollow corners, which change nothing here. Measure a clean sawn end, not the factory end, and write the four numbers on the offcut: it is the test piece for the feet in step 4.

2 Cut and label

31 pieces from 17 sticks of 2000. Label both ends of every piece.

11× | 1317 | 479 | drop 198

1317: F0 F1 F2 F3 F4 B0 B1 B2 B3 B4 S1 479: L0 L1 L2 L3 L4 R0 R1 R2 R3 R4 M0

3× | 1829 | 123 | drop 42

1829: FL FR BL 123: M1 M2 M3

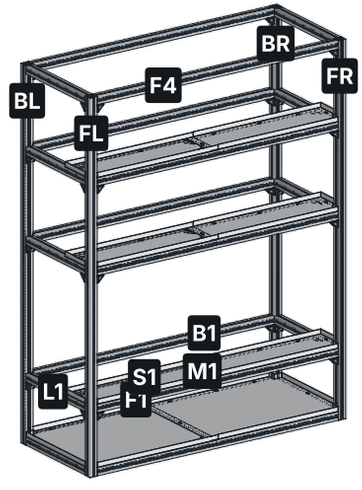
2× | 1317 | drop 680

1317: S2 S3

1× | 1829 | drop 168

1829: BR

F front, B back, L left, R right, S shelf (the tray's back edge), M mid (under the sheet laps), posts by corner. The number is the level: 0 the floor, 4 the top.

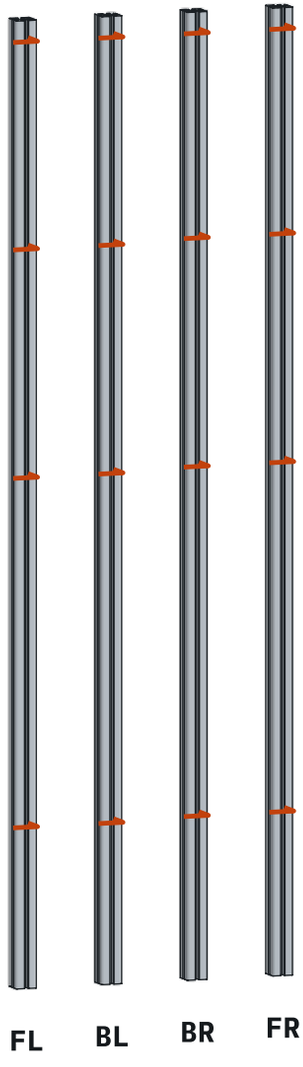


`mitre saw · square`

Bought cut to length? Check every piece against this list instead. Otherwise cut all pieces of one length in a single setup against a stop, square to 0.5, and deburr both ends: a raised burr on an end stops a rail seating square on its post.

3 Mark the levels

On all four posts, from the BOTTOM end, squared all the way round.



rail bottom, from the post's bottom end

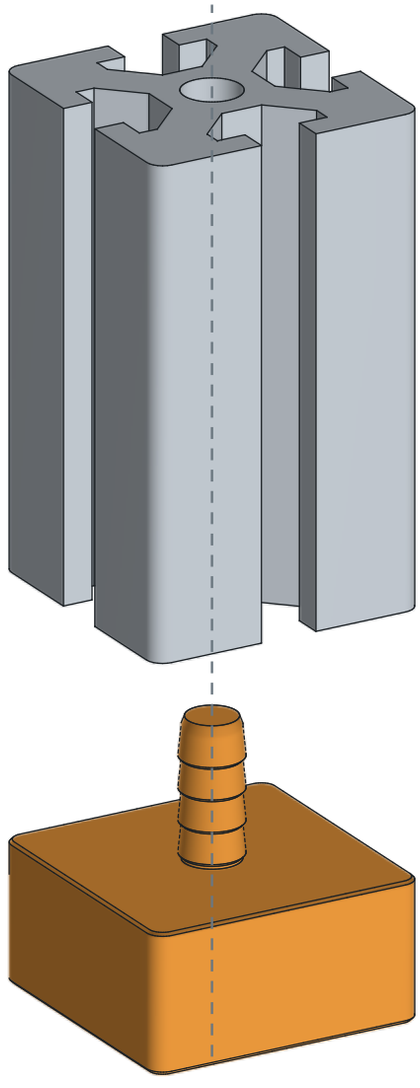
	FL, BL	FR, BR
level 4 top: dead level, the X rail hangs here	1789.0	**1789.0**
level 3 tier C	1397.0	**1402.0**
level 2 tier B	965.2	**970.2**
level 1 tier A	304.8	**309.8**
level 0 the floor: packers, step 7	0	**5.0**

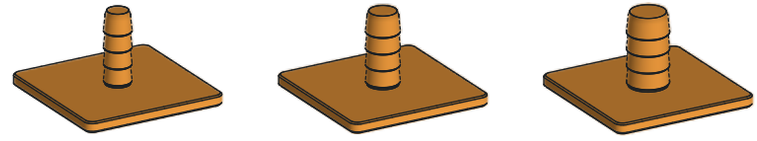
`tape · square · fine marker`

The mark is the BOTTOM face of the rail, which is the top of the gusset that carries it. The two right-hand posts are marked 5 higher on levels 0 to 3: every tray falls 5 mm to its drain at the left end (BOM G090), the pan to its leak sensor. Level 4 does not fall. Use one tape for all four posts and hook it on the same end each time: a 1 mm slip between posts is a twisted tier.

4 Print the feet

Fit-test coupons first, then four feet. SUNLU TPU 95A, pin up, no supports.





bore 6.8 **bore 8.4** **bore 10.5**

3× Fit-test coupon, one per bore

4× Foot 40 × 40 × 20, 20 mm barbed pin

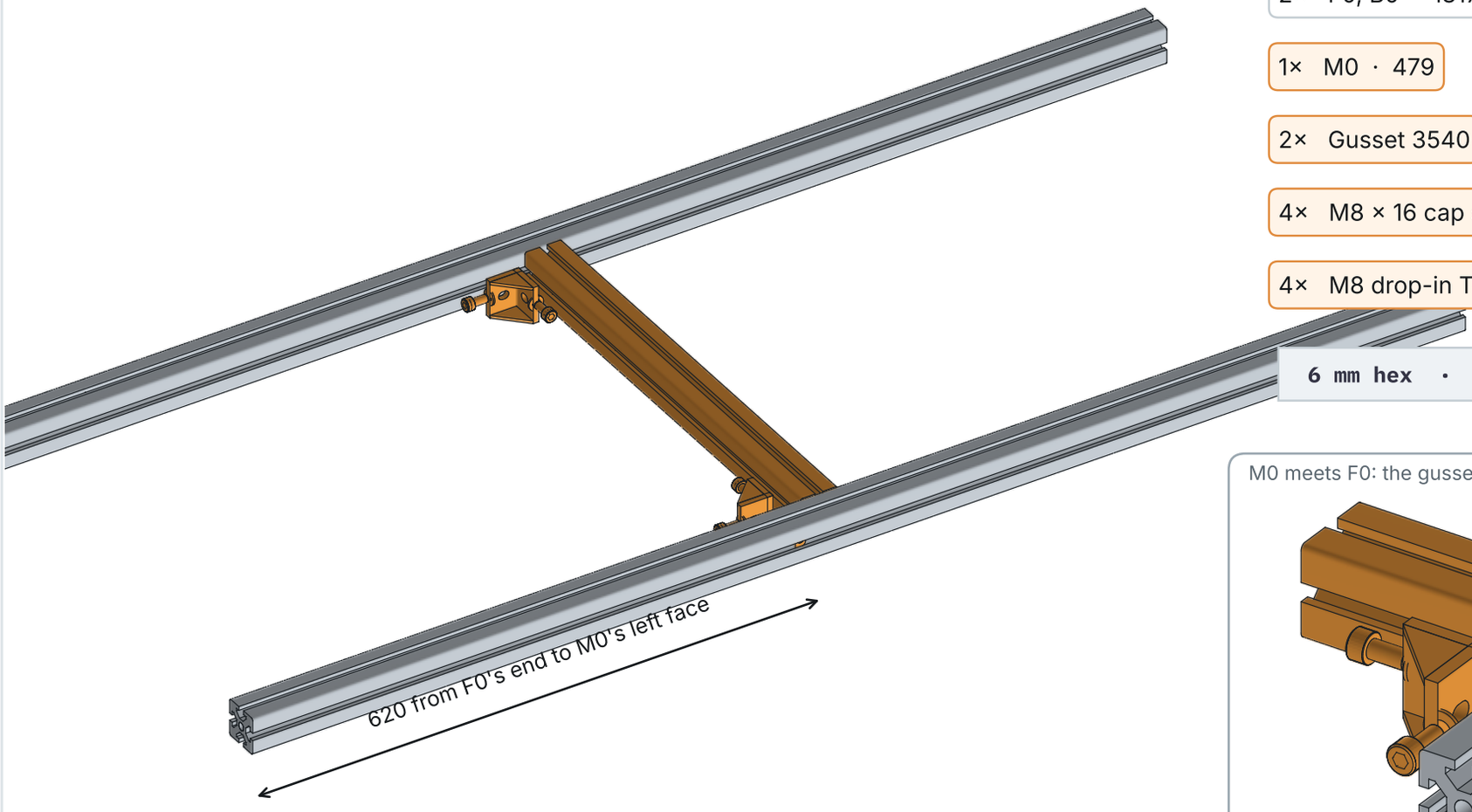
Material	SUNLU TPU 95A Clear
Infill	100 %, so a foot cannot creep
Speed	20 to 25 mm/s, retraction short
Orientation	flat face down, pin up
Each foot	about 40 g, so 160 g for four

`calipers`

Push each coupon into the bore of the offcut from step 1. Keep the one that needs a firm push and will not pull out by hand; print the four feet in that bore (tpu_foot.py --profile 4040 --bore D). Loose: raise BARB_INTERF to 0.6 or 0.7. Too tight to start: lower it. The feet do not adjust: check the floor where the cabinet will stand with the level now, and plan a shim, not a softer foot.

5 The base · front, back and the mid rail

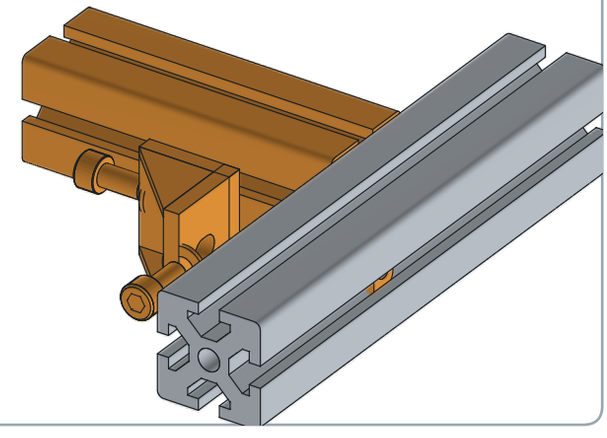
On a flat floor. The mid rail sits under the base pan's lap.



- 2× F0, B0 · 1317
- 1× M0 · 479
- 2× Gusset 3540
- 4× M8 × 16 cap screw + washer
- 4× M8 drop-in T-nut

6 mm hex · tape

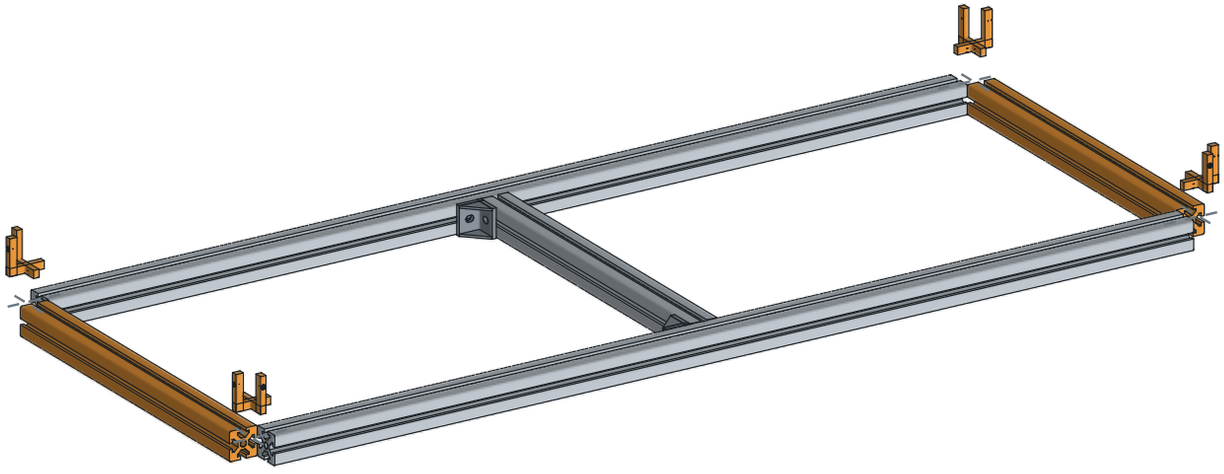
M0 meets F0: the gusset on M0's left



Lay F0 and B0 parallel, top slots up, 479 apart inside. M0 goes between them with its LEFT face 620 from F0's left end: its centre line, 680 from the frame's left edge, is where the pan's lap bolt lands. Drop each T-nut into its slot, turn it a quarter, and snug the screws: final tightening comes at step 10, once the box is square.

6 Hidden connectors in the eight floor ends

One leg into each rail's TOP slot from its end, the other leg standing up.



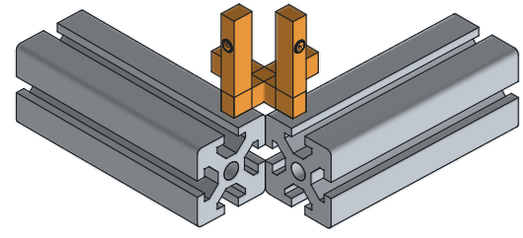
2× L0, R0 · 479

8× Hidden connector, TOCLUX

16× M6 grub, finger tight

3 mm hex

the FL corner: two connectors, two rails

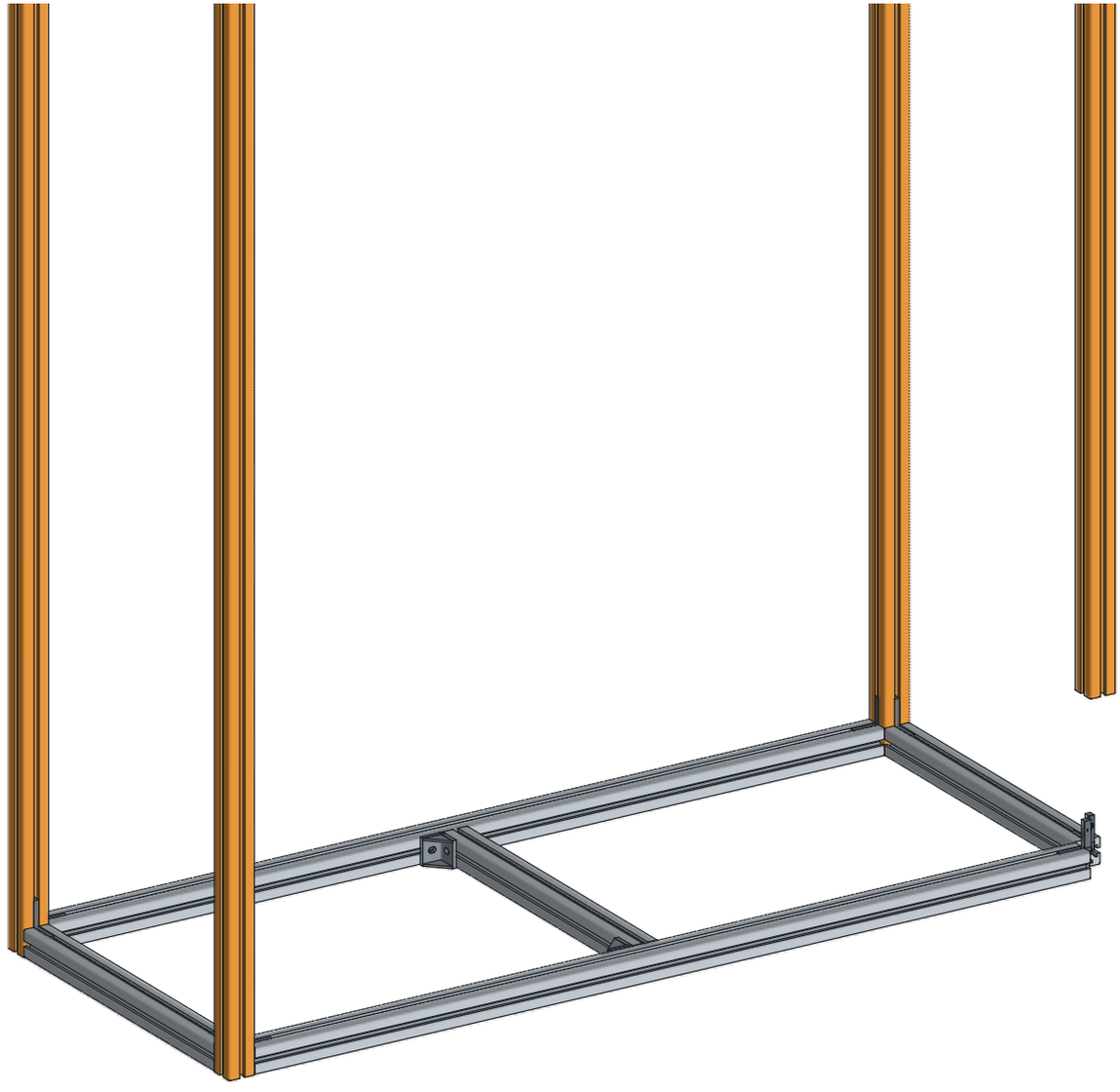


Slide each connector into the rail's top slot from the end until its upright leg stands 11 clear of the end face: that much of it sits inside the post's slot. Lay L0 and R0 in line with the ends of F0 and B0, leaving a 40 × 40 square empty at each corner for a post.

Grubs finger tight only: they lock at step 7. The connectors live inside the slots, so the base pan will lie flat over them.

7 Stand the posts

Lower each post straight down onto its two connector legs. A helper holds it plumb.

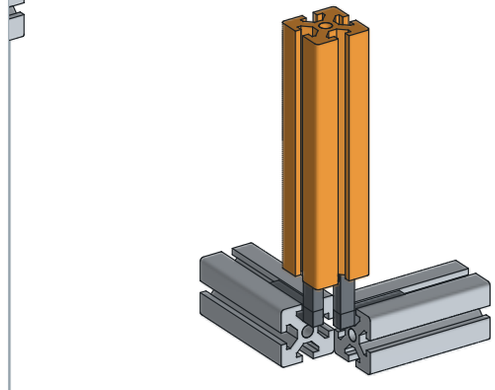


4× Post 1829, FL FR BL BR

3× 5 mm packer, base, right end

`3 mm hex · level`

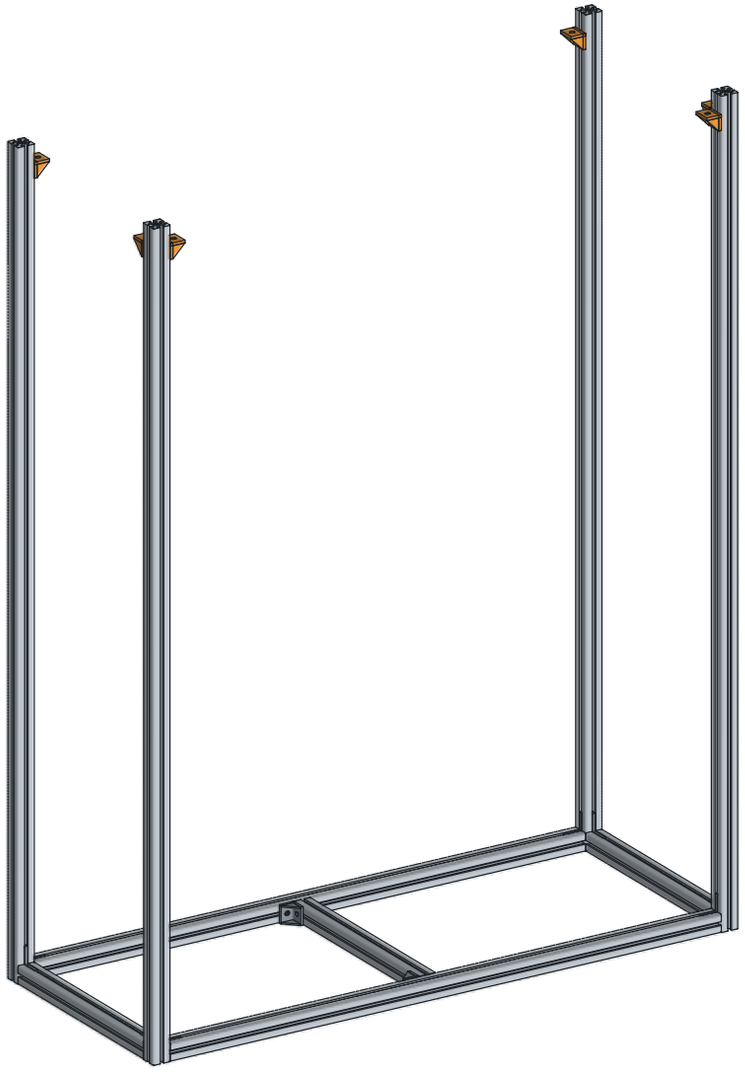
the legs enter the post's two inner slots



Before the right-hand posts go on, prop R0 and the right ends of F0 and B0 on 5 mm packers: that is the base's fall to the left (step 3). Lower the post until it stands on the floor, plumb it both ways and lock both connectors' grubs, two each. Until step 9 the posts are only held at the foot: clamp a batten across the tops, or keep the helper.

8 Ledges for the top

Six gussets on the level 4 marks. Two hidden connectors into the top side rails.

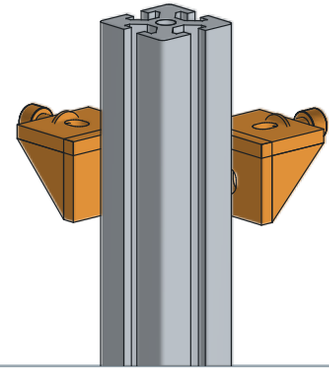


6× Gusset 3540

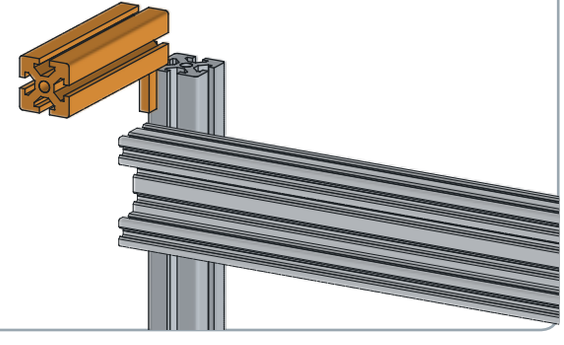
2× Hidden connector

6 mm hex · 3 mm hex

FL, level 4: one ledge per rail, gusset top on the mark



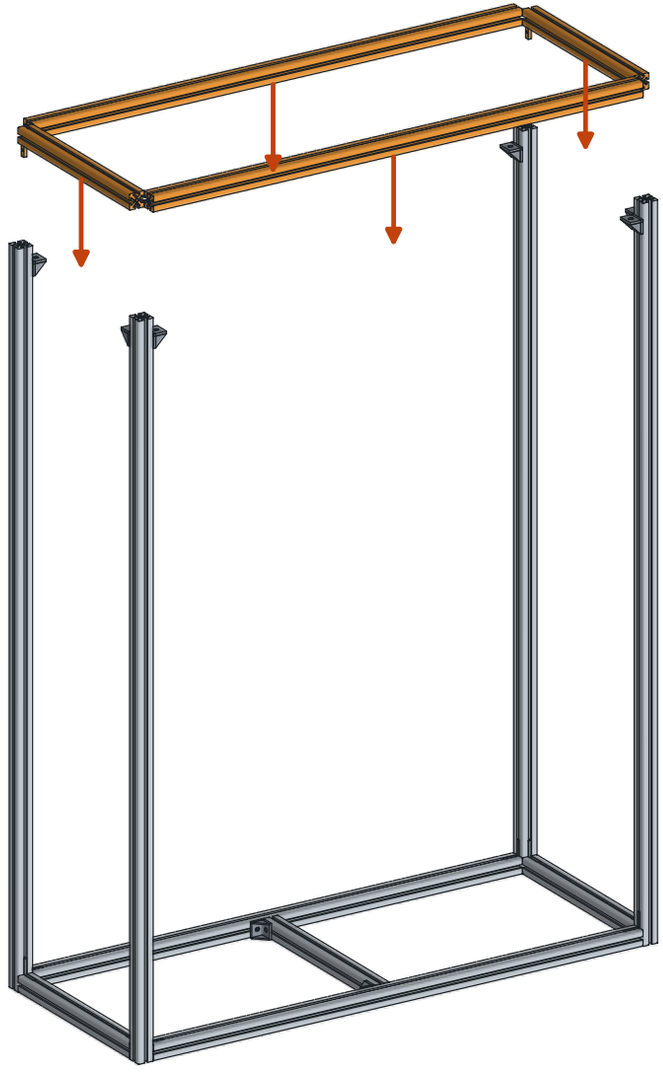
L4's back end: the X rail (pale) runs below it



Gusset TOP edge on the mark, its leg down the post; snug, then check with the square that the upper leg is level. At the two back corners the side rails get no gusset: the X rail's ends run in front of BL and BR right under them. Slide a hidden connector into the BOTTOM slot of L4 and of R4 at the back end, leg pointing down, grubs loose.

9 The top ring

Lift each rail onto its ledges. The back of L4 and R4 slides down into BL and BR.



2× F4, B4 · 1317

2× L4, R4 · 479

6× M8 × 16 cap screw + washer + T-nut

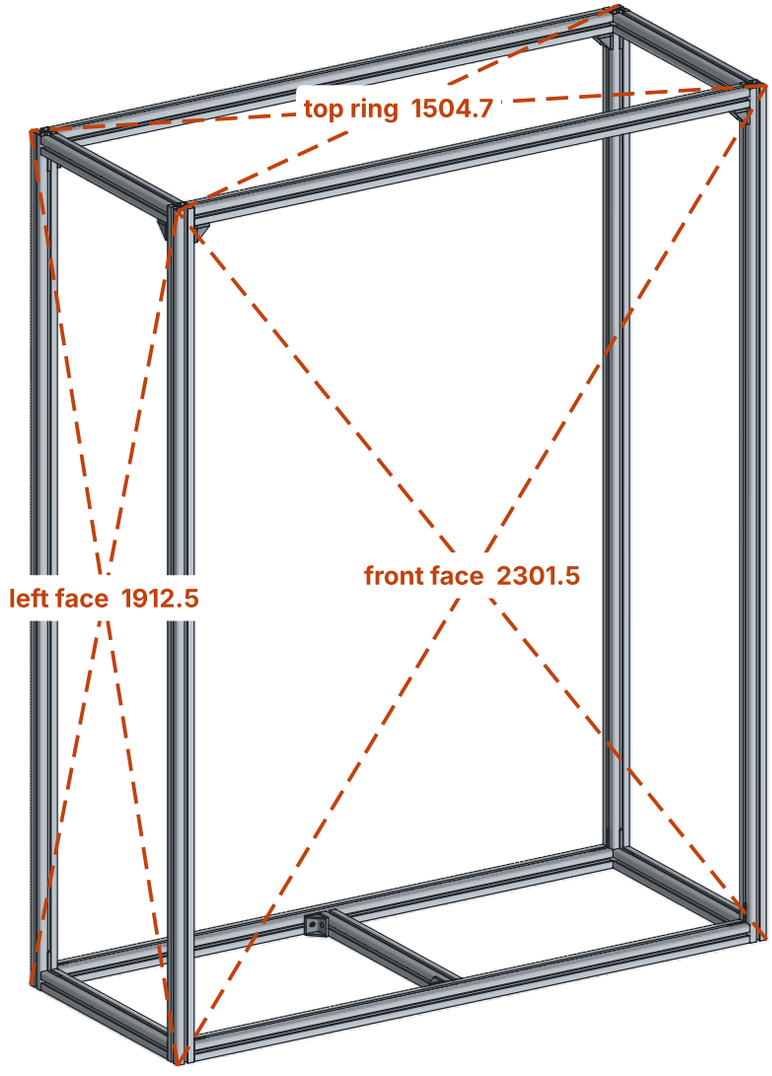
6 mm hex · 2 clamps

One screw per gusset goes in now, the upper leg into the rail's bottom slot. The lower leg went on at step 8.

F4 and B4 first: clamp each to its ledges and drop a T-nut under each gusset's upper leg. Then L4 and R4: front end on its gusset, back end lowered so the connector's leg slides down the post's front slot. Lock those grubs. Now the box stands on its own: the batten can come off.

10 Square it, then tighten

Diagonals equal, posts plumb. Only then the final turn on every screw so far.



Diagonals

within 2 of each other, every face

Posts

plumb both ways, on two faces each

Top ring

dead level both ways: the gantry hangs from it

Then

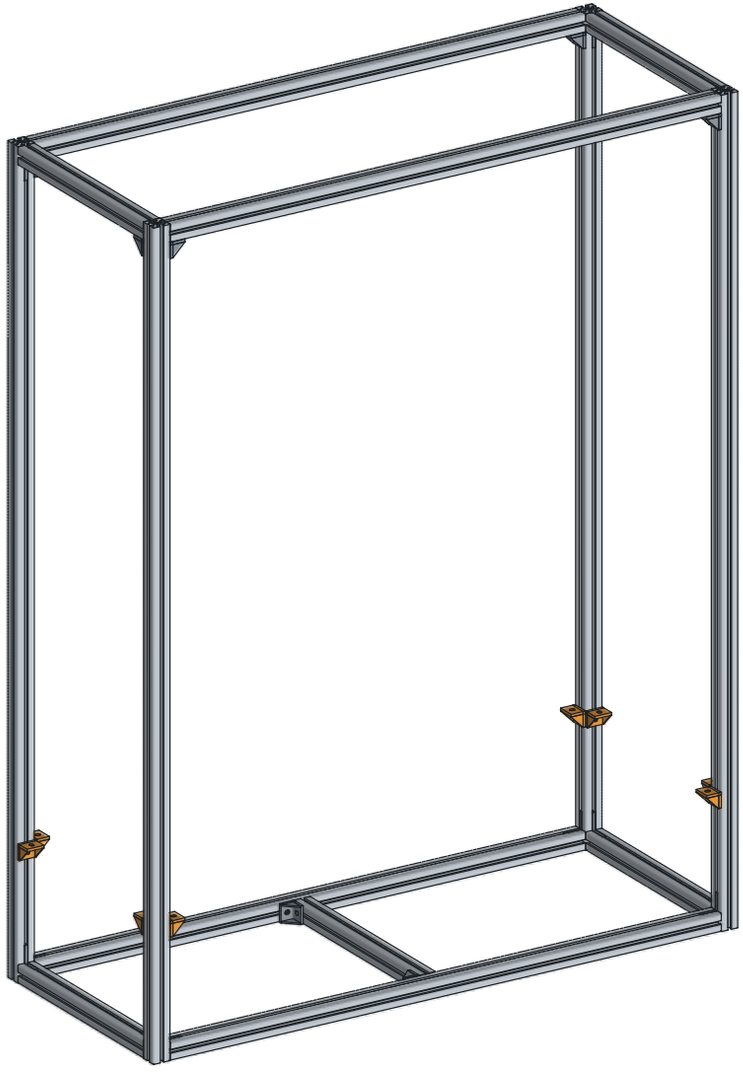
tighten every gusset screw so far

`tape · level · 6 mm hex`

Measure corner to corner on the outside of the posts, the tape hooked the same way both times. The base ring's diagonals are the same as the top's. Long diagonal long: push that corner in, gently, with the screws still snug. A cabinet that starts out of square stays out of square, and the trays will not drop in at step 19. The right-hand base packers stay in until the feet go on at step 15.

11 Tier A · eight ledges

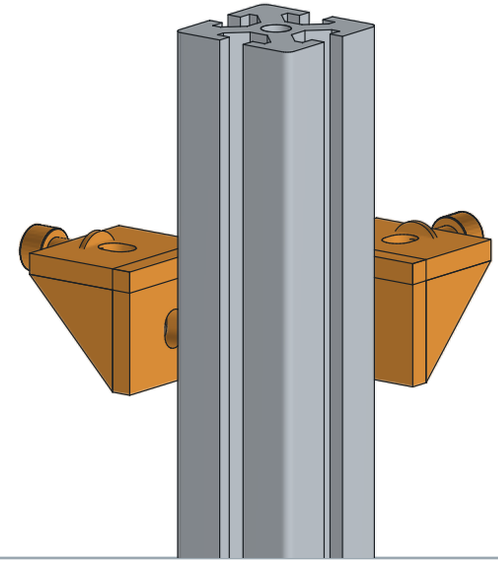
On the level 1 marks: 304.8 on the left posts, 309.8 on the right.



8× Gusset 3540

8× M8 × 16 + washer + T-nut, lower legs

FL: a ledge for F1, a ledge for L1



6 mm hex · square

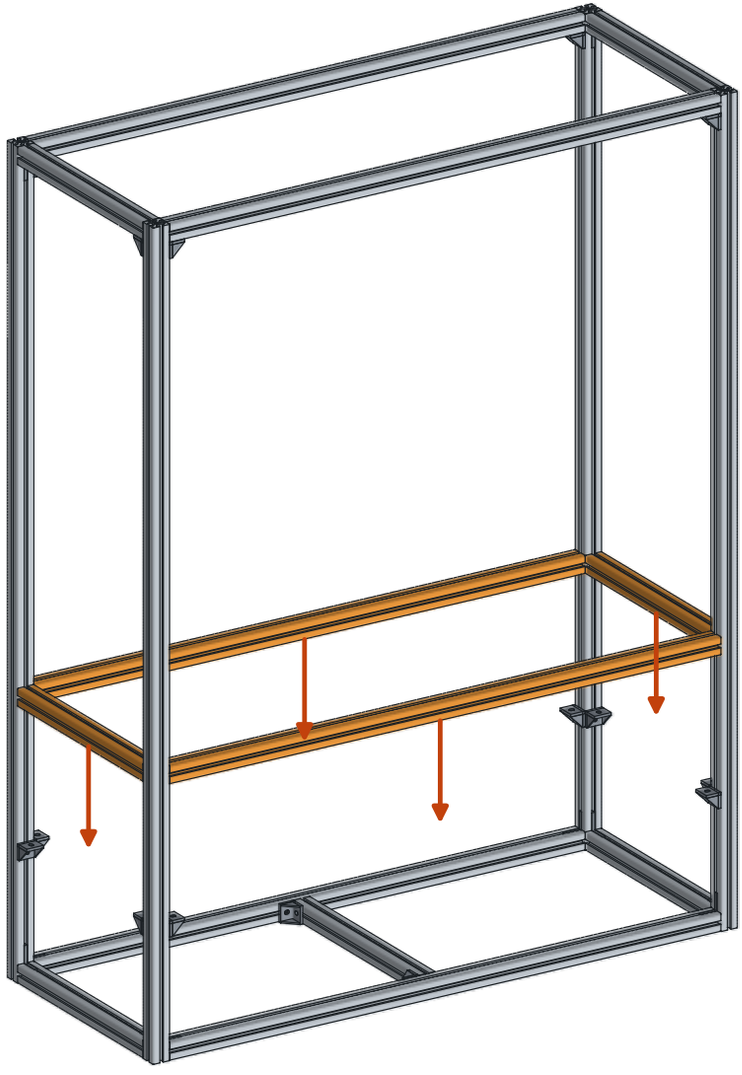


where this is

Same as step 8: TOP edge of each gusset on the mark, lower leg down the post, one screw. Two ledges at every post, on the two inner faces. The 5 mm between the left and right marks is the tray's fall to its drain: do not level it out.

12 Tier A · front, back and side rails

Onto the ledges, clamp, one screw up through each gusset into the rail.



2× F1, B1 · 1317

2× L1, R1 · 479

8× M8 × 16 + washer + T-nut, upper legs

6 mm hex · 2 clamps

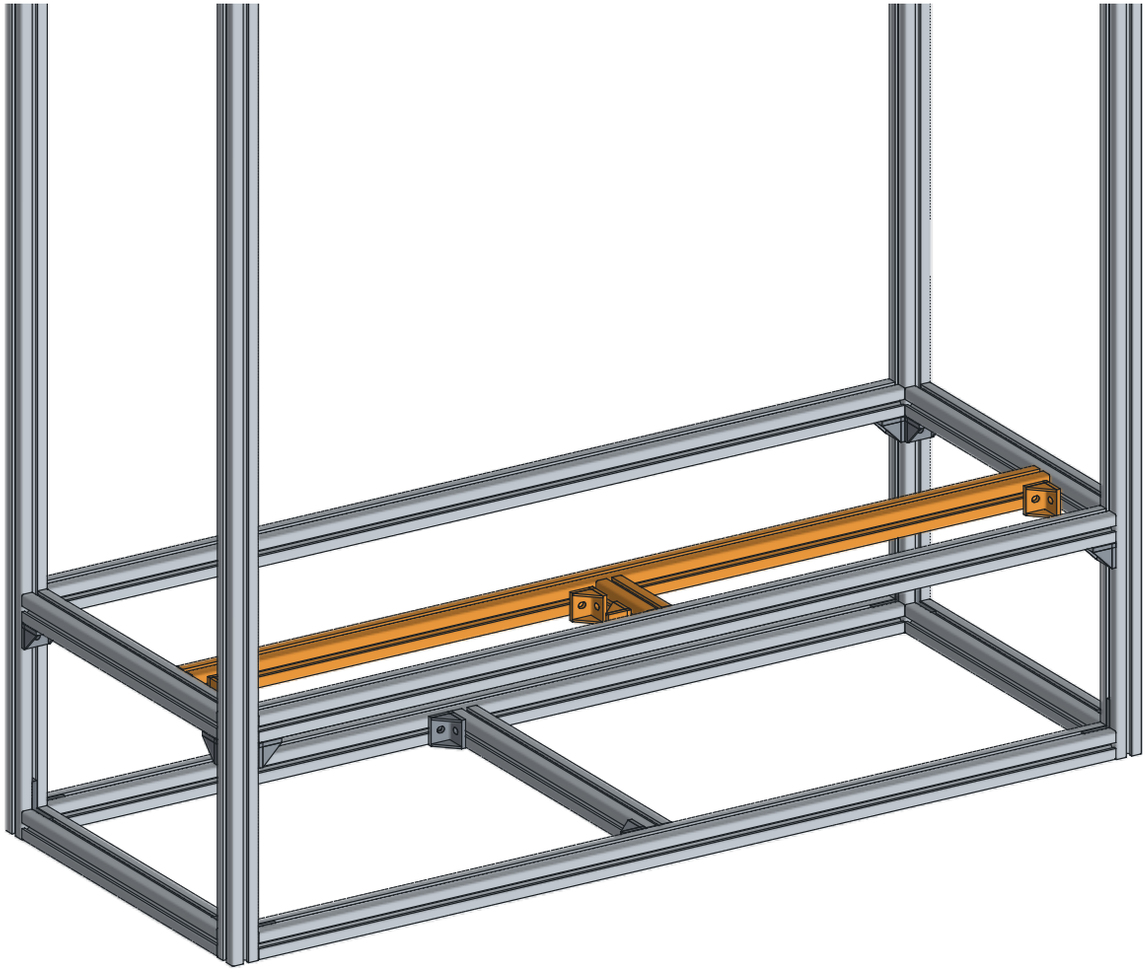


where this is

Each rail sits on two ledges and butts the posts at both ends. Push it hard against the post while you snug the screw, so the end face seats square. Gaps at the ends mean the posts have spread: loosen the ledges, close the gap, re-check the diagonals.

13 Tier A · the tray rails

S1 carries the tray's back edge, M1 its lap. Four gussets, all on the tray side.



1× S1 · 1317

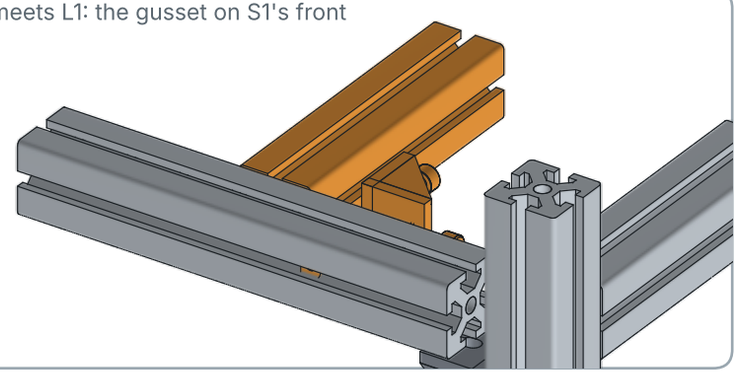
1× M1 · 123

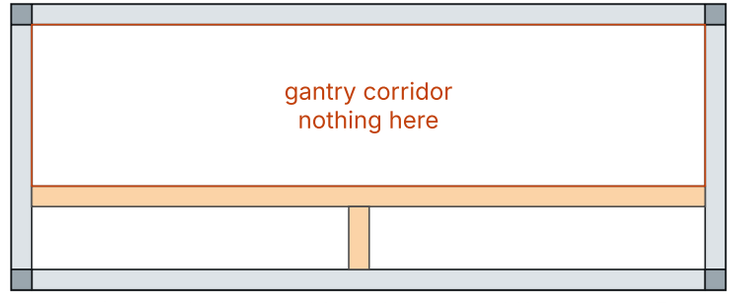
4× Gusset 3540

8× M8 × 16 + washer + T-nut

6 mm hex · tape

S1 meets L1: the gusset on S1's front



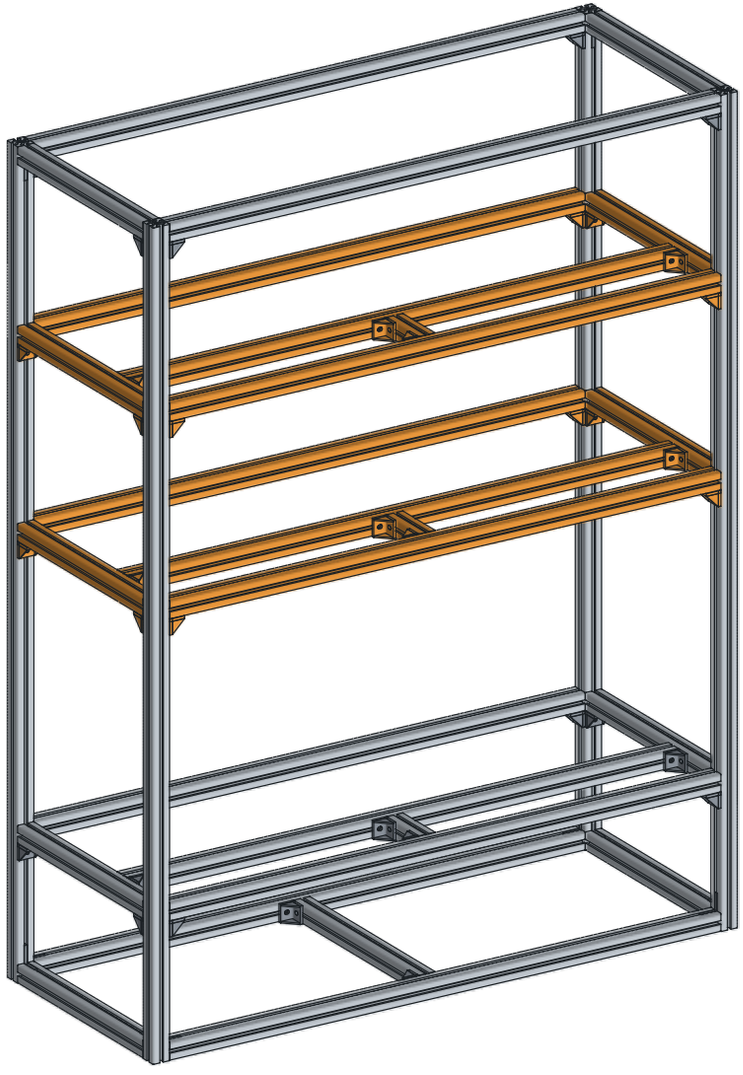


plan, front at the bottom: S1 at y 163 to 203, M1 at x 660 to 700

S1's back face lands on the tray line, 203 from the front: measure it at both ends. These four gussets stand on edge in the inside corners, on the FRONT side of S1 and the LEFT side of M1. One on the back of S1 would sit in the gantry's corridor, which must stay empty from 203 back to the posts. Bolt S1 to L1 and R1 first, then fit M1 between F1 and S1, its left face 620 from F1's end.

14 Tiers B and C · the same three steps

Repeat 11 to 13 on the level 2 and level 3 marks.



Tier B, level 2

965.2 left posts, 970.2 right

Tier C, level 3

1397.0 left posts, 1402.0 right

- 6× F, B, S · 1317
- 4× L, R · 479
- 2× M · 123
- 24× Gusset 3540
- 48× M8 × 16 + washer + T-nut

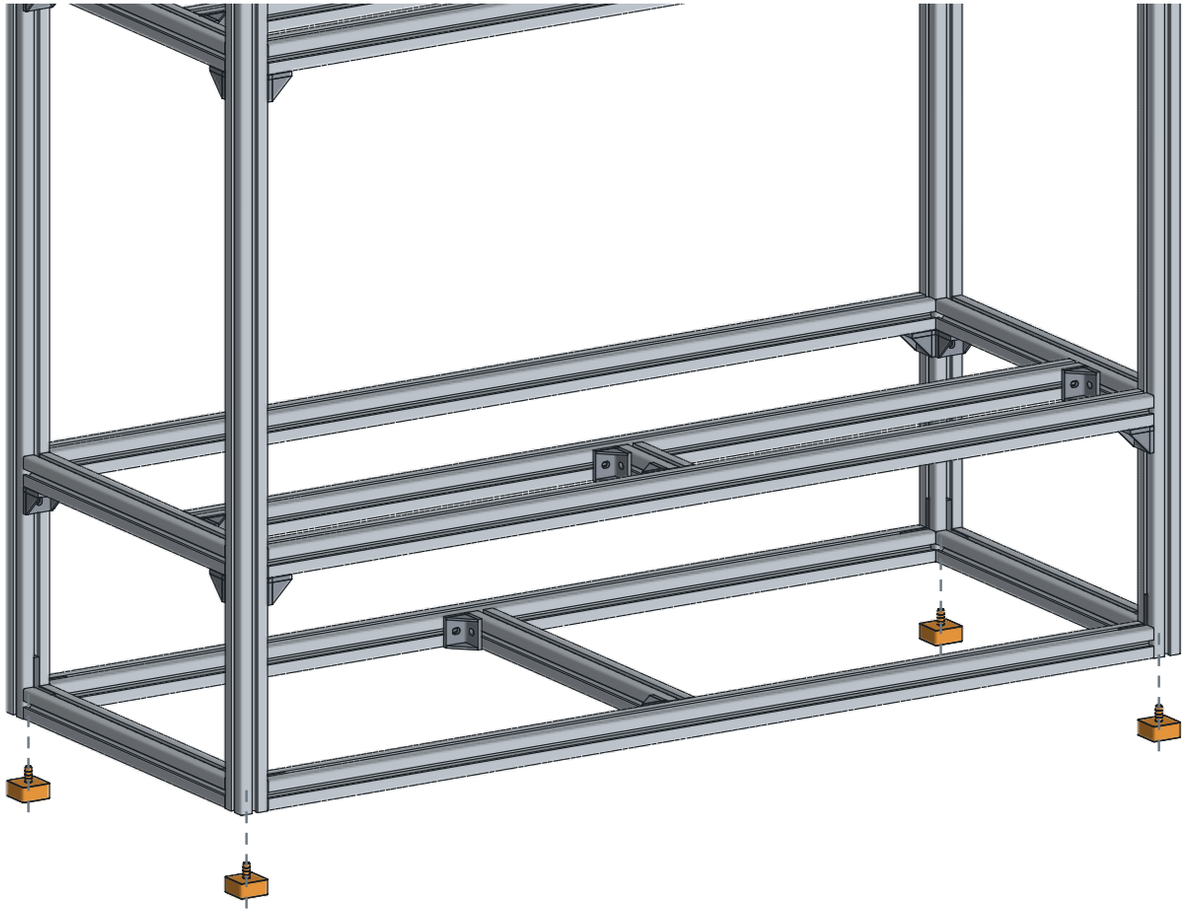


where this is

Work bottom up so you are never reaching through a finished tier. When tier C is in, re-check the diagonals, then tighten every gusset screw in the cabinet. 44 gussets, 88 screws: tick them off on the cut list as you go.

15 The feet

One corner at a time: lift 25, push the foot's pin up into the post, lower.

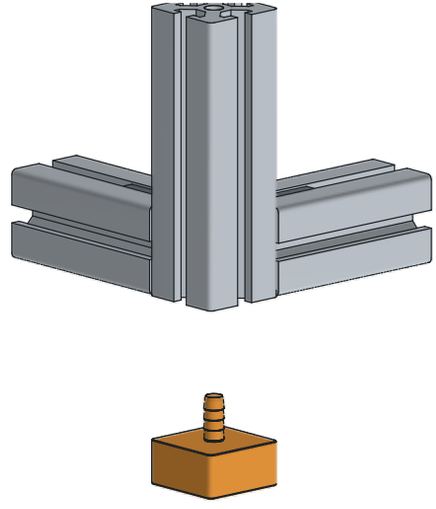


4× Foot, printed in step 4

bar and a block

Pull the 5 mm base packers out as each right-hand corner comes up.

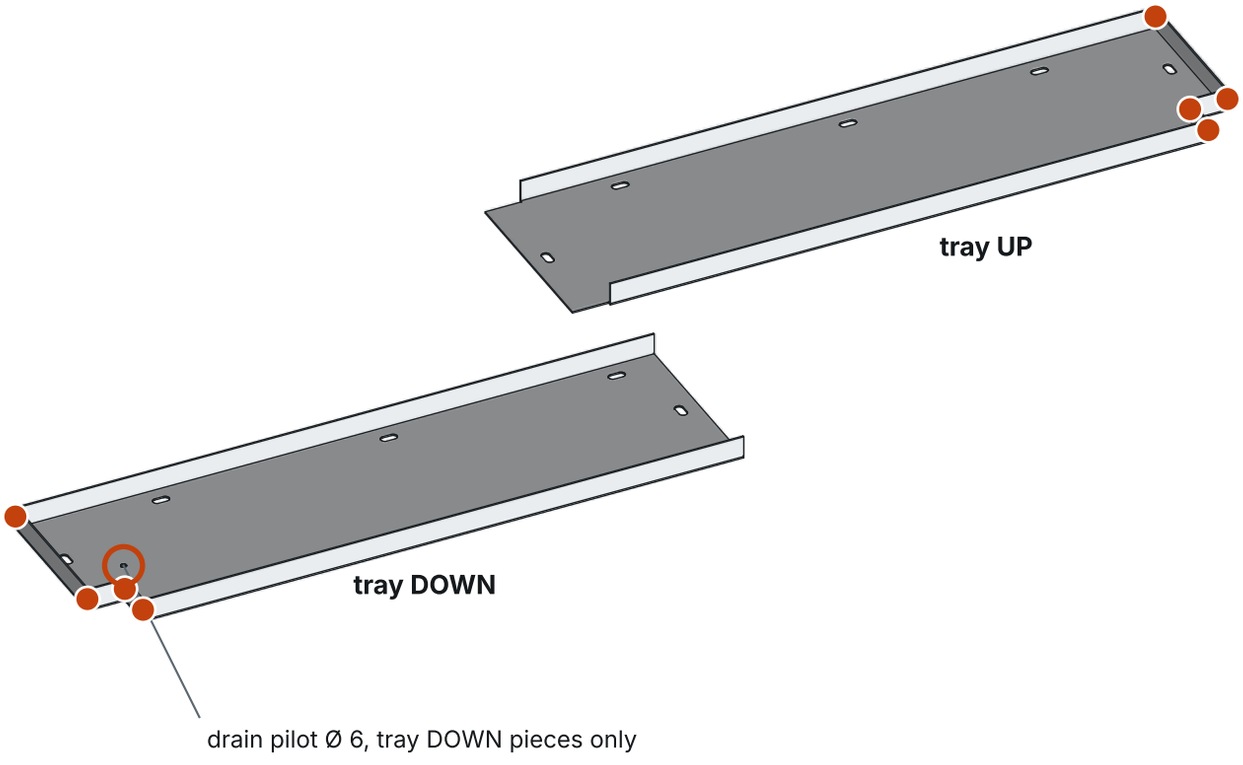
FL lifted 25: the pin goes up into the bore



Lift no more than 25: the corner only has to clear the pin. Line the foot up square with the post, push until its top face meets the post's end all round, then lower it onto the foot. Do the diagonal corner next, then the other two. The whole cabinet now stands 20 higher: everything above is measured from the bottom of the posts, so none of the marks move.

16 The sheets · check, drill, seal

Eight pieces from SendCutSend. Drain holes opened, every corner sealed, before any goes in.



Check off the order

3×	**tray DOWN**	700 × 203, notch at the left
3×	**tray UP**	737 × 203, tongue at the left
1×	**base pan DOWN**	700 × 559, two notches
1×	**base pan UP**	737 × 559, two notches

● a bead of silicone, inside

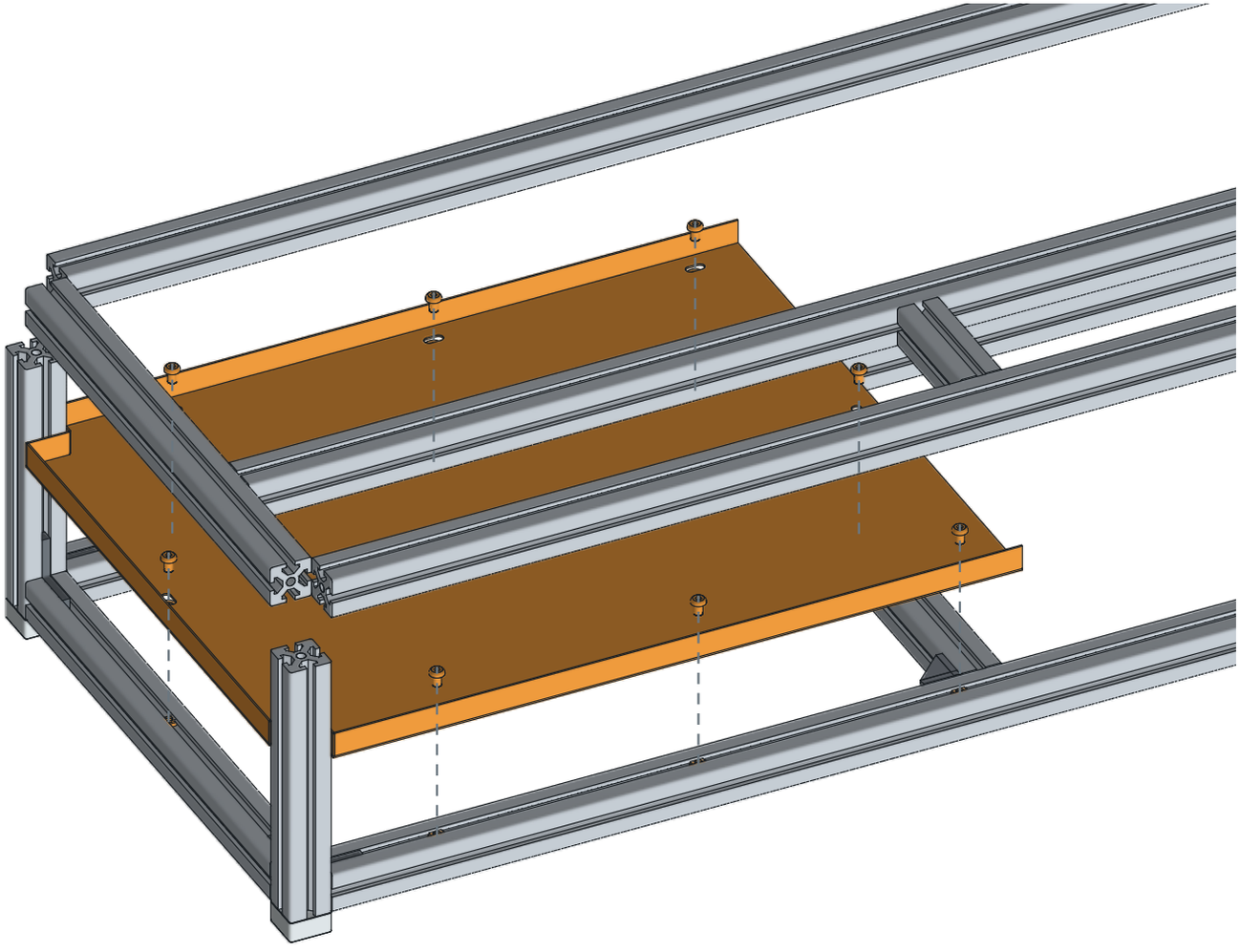
`step drill · caulking gun`

1× Neutral-cure silicone

Every bend 90° and every lip 25 tall: a soft bend lifts the floor off its rails. Step-drill the Ø 6 pilot in each of the three tray DOWN pieces to the bulkhead's own hole size (on its packet, usually about 22) and deburr both faces. Then a bead of neutral-cure silicone into every inside corner where two lips meet, tooled smooth: the mitres there are cut 2.5 open. Let it skin over before the sheets go in.

17 Base pan · T-nuts, then the DOWN piece

Eight drop-in nuts under the DOWN piece's holes, the piece on top, seven screws.



- 1× Base pan, DOWN piece
- 8× M8 drop-in T-nut
- 7× M8 × 12 button head + washer

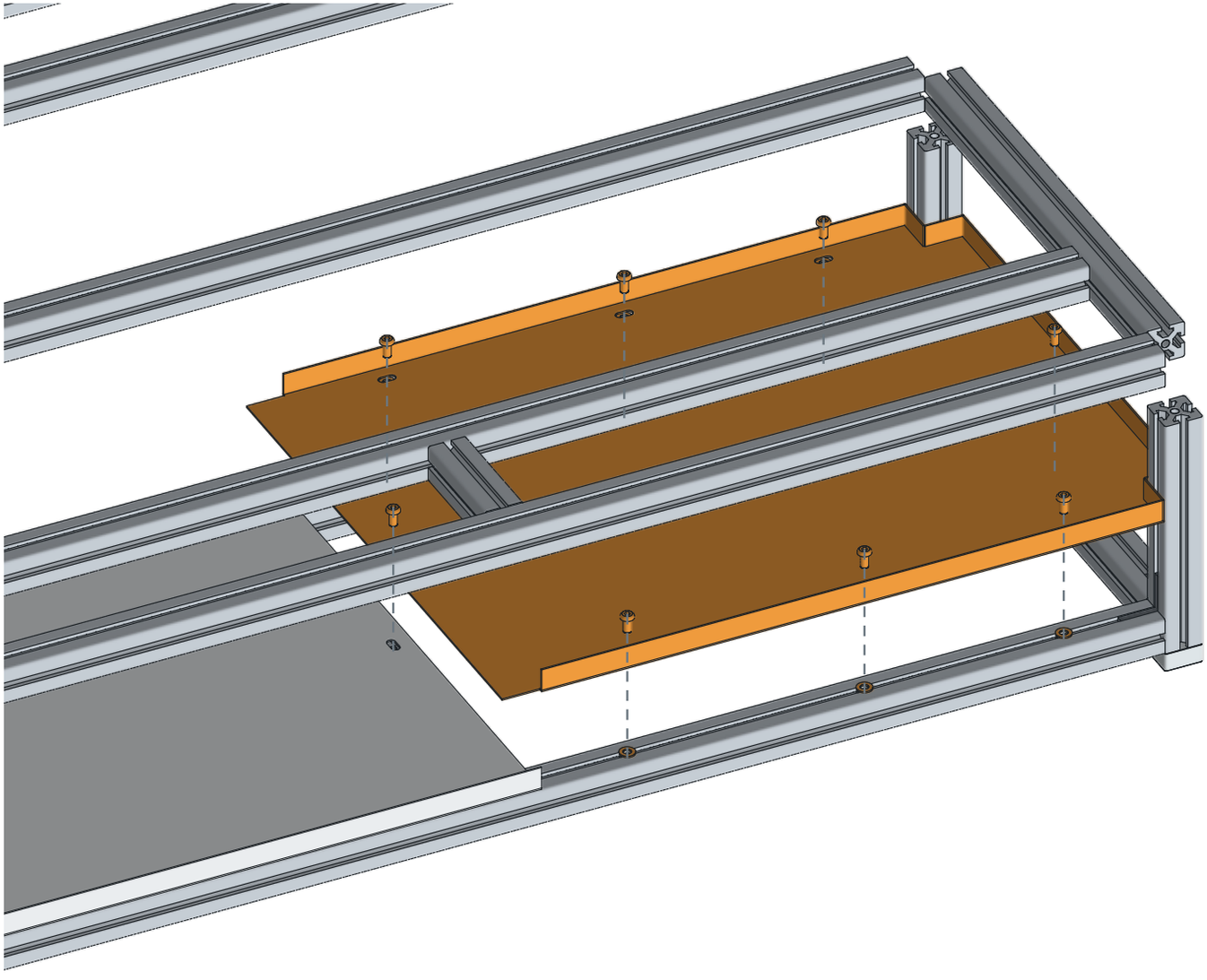
5 mm hex

The eighth hole, on M0 at x 680, is the lap: its nut goes in now, its screw at step 18, through both pieces.

Drop a nut into the top slot under each hole, a quarter turn to lock it, then slide it along until it sits under the hole. Lay the DOWN piece on with its notches round FL and BL, its lap end on M0. Start all seven screws before tightening any, then snug them from the middle outward. The slotted holes take up 4 mm either way: centre the piece in them, do not force it.

18 Base pan · the UP piece over the lap

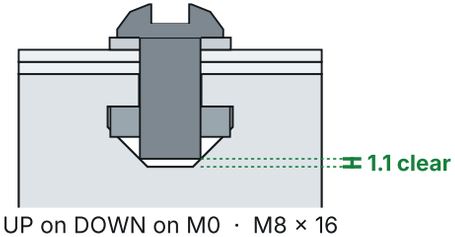
A shim on every one of its seven slots, a bead along the lap, then the piece, eight screws.



- 1× Base pan, UP piece
- 7× M8 washer as a shim, under the sheet
- 8× M8 × 16 button head + washer

5 mm hex · caulking gun

the lap at x 680, in section

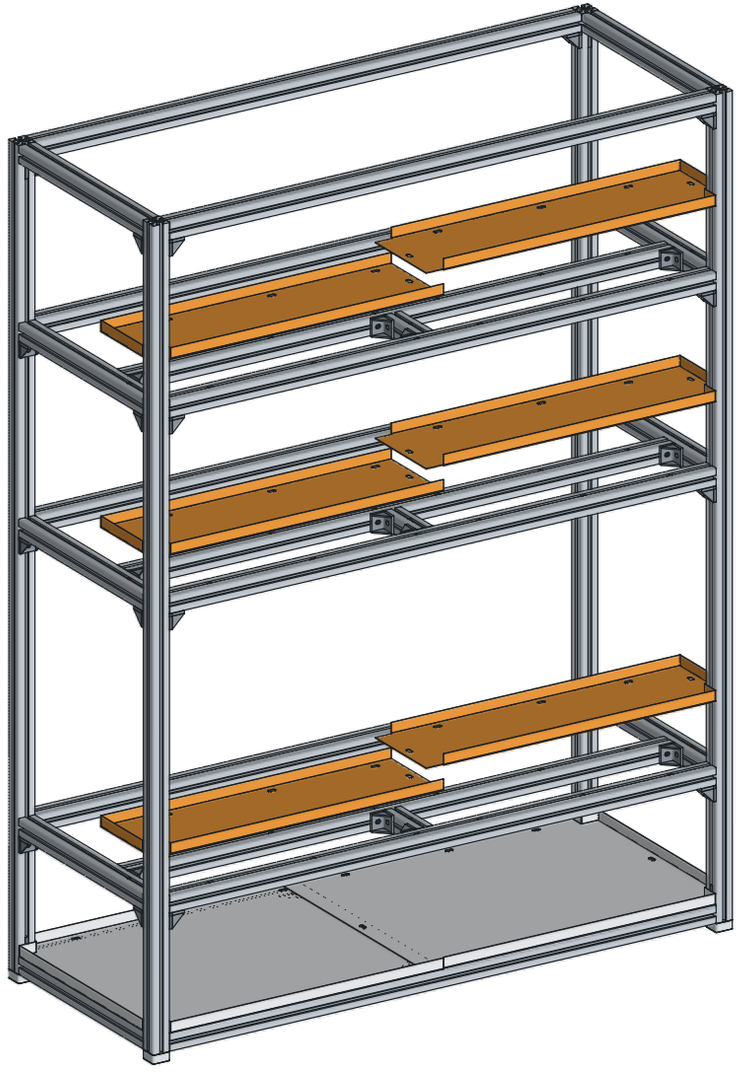


UP on DOWN on M0 · M8 × 16

The UP piece lies on the DOWN piece over the 40 lap, so the rest of it stands 1.6 proud of its rails: a washer under each of its other seven slots fills that, or the screw pulls the sheet down into a dent. Run a bead of silicone along the DOWN piece 10 in from its edge before the UP piece goes down. The UP piece is on top because water runs left, down the fall: it sheds over the lap, never under it.

19 Trays · the same, three times

Steps 17 and 18 on tiers A, B and C. 203 deep: the tray stops at the tray line.



Each tray

8 nuts, DOWN piece and 7 × M8 × 12,
7 shims, UP piece and 8 × M8 × 16

All three

24 nuts, 21 × M8 × 12, 21 shims,
24 × M8 × 16

Check

the tray's back lip on S's back face,
flush with the tray line at 203

3× Tray, DOWN piece

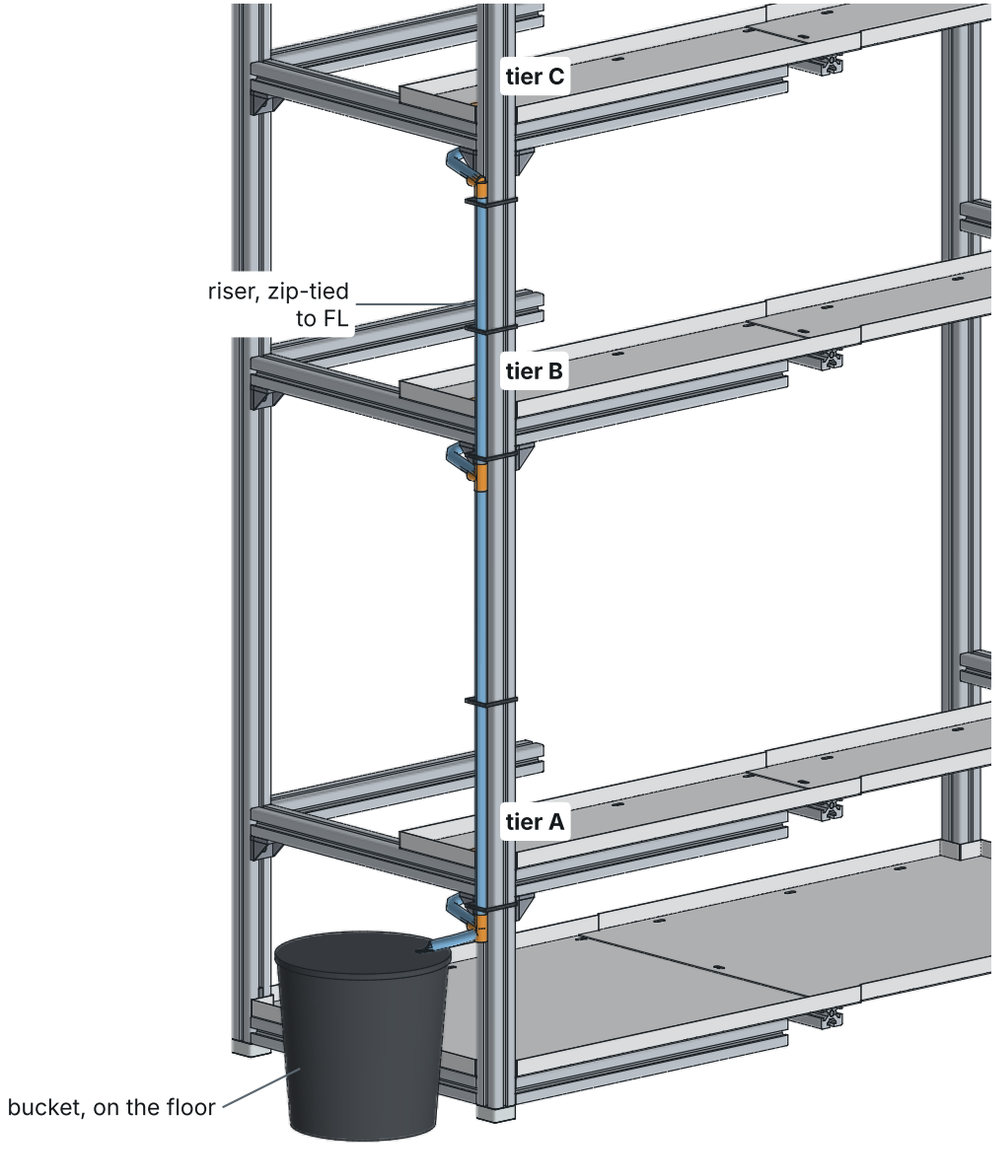
3× Tray, UP piece

5 mm hex · caulking gun

Bottom tier first, so nothing you drop lands in a finished tray. Each tray hugs FL and FR with its 41 × 41 notches and stops at the tray line: its back lip must not lean into the corridor. If a tray will not sit flat, a ledge is off its mark: fix the frame, not the tray.

20 Drains · one riser on the FL post

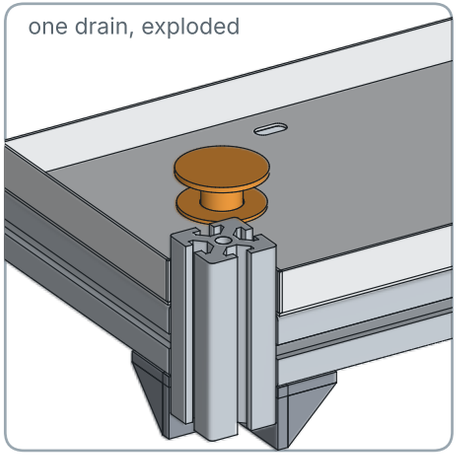
A bulkhead in each tray, a short branch out of the side, one riser zip-tied to the post.

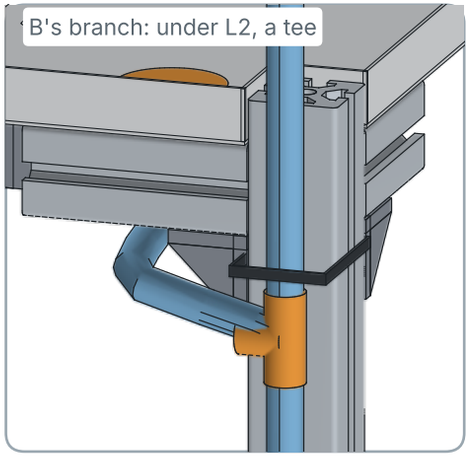


- 3× Bulkhead 1/2 in, EPDM washers
- 3× Elbow 90°, 1/2 in MNPT × 1/2 in barb
- 1× Barbed elbow 1/2 in
- 2× Barbed tee 1/2 in
- 1× Drain tube 1/2 in ID, 10 ft
- 5× Zip tie 300
- 1× Bucket, 2 gal, lidded

BOM G090 to G090c.

C	elbow at the top of the riser
B, A	a tee each, 40 under the rail
Pan	no drain: the dry catch tray


one drain, exploded


B's branch: under L2, a tee

Flange and one washer inside the tray, the second washer and the nut underneath; hand tight plus a quarter turn. Screw an elbow into each bulkhead, run its branch under the left rail and out of the side, falling all the way. Zip-tie the riser to FL every 300 or so, never kinked, and lead its end through a hole in the bucket lid. Every tray drains on its own: no tray takes another's runoff. Never plumb it back to the reservoir.

21

Before you load it

The checks that catch the usual faults, while the cabinet is still empty.

- [ ] **Anchored to the wall**
  Two anti-tip brackets into studs, BOM G015. 1.85 m tall and 0.56 deep tips forward with the trays full.

- [ ] **Every screw tight**
  88 gusset screws, 60 sheet screws, 20 grubs. A paint mark on each as you go.

- [ ] **The falls**
  A level on each tray floor: the bubble sits toward the right. Water stands nowhere.

- [ ] **Water test**
  1 L into each tray. Each one empties down its branch and the riser into the bucket. The pan stays dry.

- [ ] **The corridor**
  A tape run along the tray line at 203: nothing of the frame behind it but the back posts and back rails.

- [ ] **The top**
  Level both ways across F4 and B4: the gantry's X rail hangs below B4 (Gantry Rev E, step 23).

Leak sensors: one in the base pan, one beside the bucket (BOM G074). The panels and doors are the next booklet; the gantry booklet starts from here at its step 21.